هوش مصنوعی چیست؟

- شبیه سازی توانمندی های انسان از قبیل دیدن، شنیدن، صحبت کردن، درک مفاهیم، یادگیری، تصمیم گیری و ... در ماشین ها
- در ارتباط مستقیم با علوم شناختی، علوم اعصاب، روانشناسی، زبان شناسی و ...
- **اهداف هوش مصنوعی:**
  - یادگیری
  - استدلال
  - درک مفاهیم
  - تصمیم گیری
  - پیش بینی
  - فعالیت یا اکشن که باعث تغییر محیط شود

سیستم های AIضعیف

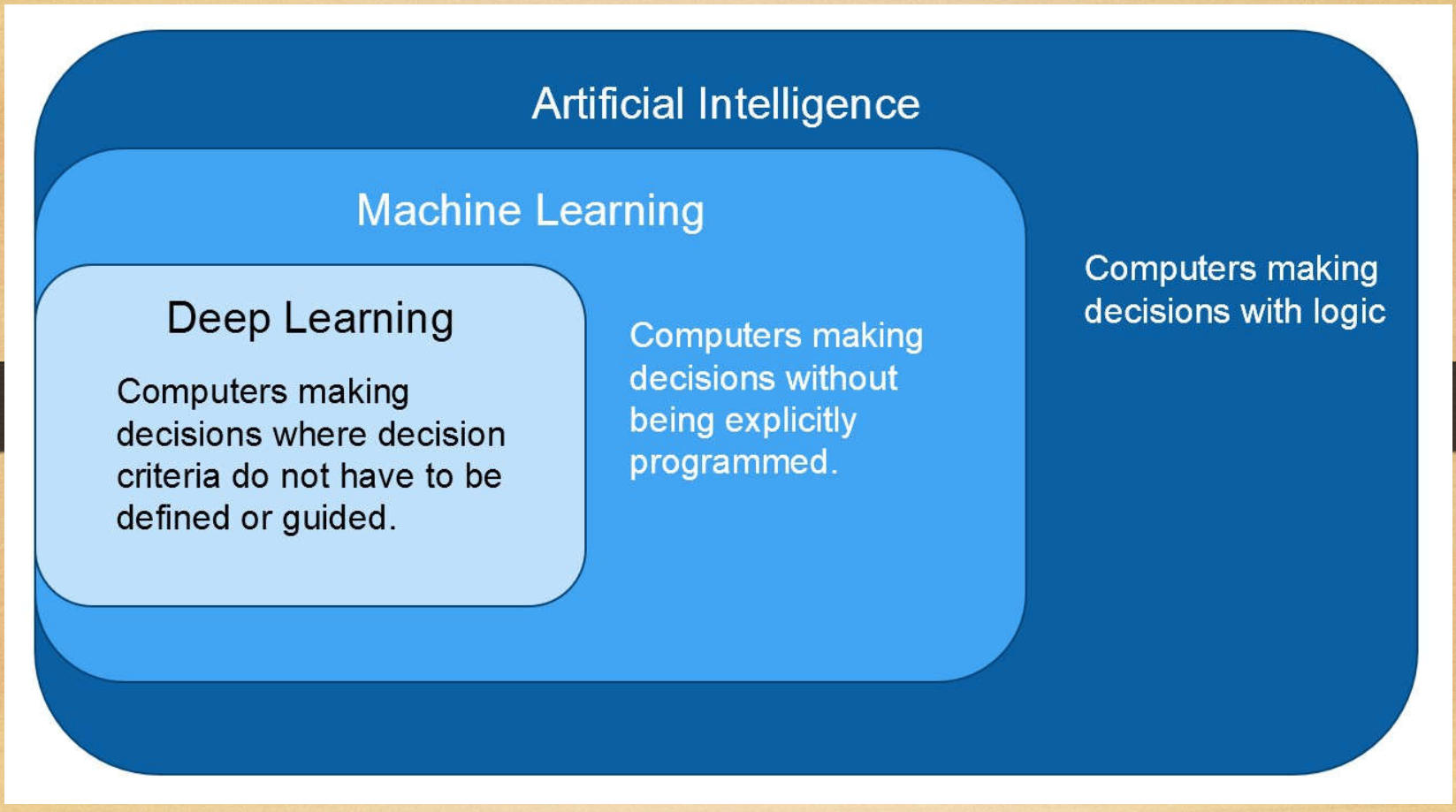
- Amazon Alexa
- Apple Siri
- بازی های ویدئویی



ابزار کمکی انسانها

سیستم های AIقوی

- خودروهای خودران
- اتاق عمل


Artificial Intelligence
Machine Learning
Deep Learning
Computers making decisions where decision criteria do not have to be defined or guided.
Computers making decisions without being explicitly programmed.
Computers making decisions with logic

یادگیری ماشین چیست؟

- Machine learning can be broadly defined as **computational methods** using experience to improve performance or to **make accurate predictions**.
- Here, *experience* refers to the past **information available** to the learner, which typically takes the form of **electronic data** collected and made available for analysis.
- This data could be in the form of **digitized human-labeled training sets**, or other types of information obtained via **interaction with the environment**.
- In all cases, its **quality** and **size** are crucial to the success of the predictions made by the learner.

یادگیری ماشین چیست؟

- An example of a learning problem is how to use a finite sample of randomly selected **documents**, each labeled with a **topic**, to accurately **predict the topic of unseen documents.**
- Clearly, the **larger is the sample**, the easier is the task.
- But the difficulty of the task also depends on the **quality of the labels** assigned to the documents in the sample, since the labels may not be all correct, and on the **number of possible topics.**

یادگیری ماشین چیست؟

- Machine learning consists of designing **efficient and accurate prediction** *algorithms*.
- As in other areas of computer science, some critical measures of the quality of these algorithms are their **time and space complexity**.
- But, in machine learning, we will need additionally a notion of ***sample complexity*** to evaluate the sample size required for the algorithm to learn a family of concepts. More generally, theoretical learning guarantees for an algorithm depend on the **complexity of the concept classes** considered and the **size of the training sample**.

یادگیری ماشین چیست؟

- Since the success of a learning algorithm depends on the data used, machine learning is inherently related to **data analysis and statistics**.
- More generally, learning techniques are **data-driven methods** combining fundamental concepts in computer science with ideas from **statistics, probability and optimization**.

کاربردهای یادگیری ماشین



کاربردها

- پزشکی و بیولوژی
- خودروهای خودران
- پردازش زبان طبیعی
- تحلیل سیگنالهای صوتی، گفتار و موسیقی
- تولید موسیقی، تولید شعر
- سیستم های مالی
- پیشنهاد دهنده ها

Generative AI

کاربردها

- تبلیغات آنلاین
- بینایی ماشین
- مخابرات
- تشخیص دست نوشته
- پیش بینی زلزله
- تحلیل شبکه های اجتماعی

سناریوهای یادگیری

- Supervised learning
- Unsupervised learning
- Semi-supervised learning
- Self-supervised learning
- Ensemble learning

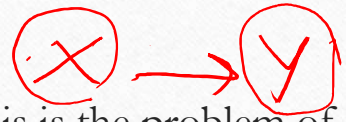


سناریوهای یادگیری

- Transfer learning
- Multi-task learning
- Adversarial learning
- Online learning
- Reinforcement learning

دسته بندی مسائل یادگیری

- ***Classification***: this is the problem of assigning a category to each item. For example, document classification consists of assigning a category such as *politics*, *business*, *sports*, or *weather* to each document, while image classification consists of assigning to each image a category such as *car*, *train*, or *plane*.



- ***Regression***: this is the problem of predicting a real value for each item. Examples of regression include prediction of stock values. In regression, the penalty for an incorrect prediction depends on the magnitude of the difference between the true and predicted values, in contrast with the classification problem, where there is typically no notion of closeness between various categories.



دسته بندی مسائل یادگیری

- ***Ranking***: this is the problem of learning to order items according to some criterion. Web search, e.g., **returning web pages relevant to a search query,** is the canonical ranking example.
- ***Clustering***: this is the problem of partitioning a set of items into homogeneous subsets. Clustering is often used to analyze very large data sets. For example, in the context of social network analysis, clustering algorithms attempt to identify natural *communities* within large groups of people.
- ***Dimensionality reduction*** **or** ***manifold learning***: this problem consists of transforming an initial representation of items into a lower-dimensional representation while preserving some properties of the initial representation. A common example involves preprocessing digital images in computer vision tasks.

A first course in probability

sheldon Ross

Murphy

Learning Stages

- *Examples*: Items or instances of data used for learning or evaluation. In our spam problem, these examples correspond to the **collection of email messages** we will use for learning and testing

Features: The set of attributes, often represented as a vector, associated to an example. In the case of email messages, some relevant features may include the **length of the message, the name of the sender, various characteristics of the header, the presence of certain keywords in the body of the message**, and so on.

Labels: Values or categories assigned to examples. In classification problems, examples are assigned specific categories, for instance, the **spam and non-spam categories** in our binary classification problem. In **regression**, items are assigned real-valued labels.

Hyperparameters: Free parameters that are not determined by the learning algorithm, but rather specified as inputs to the learning algorithm.

Learning Stages

• *Training sample*: Examples used to train a learning algorithm. In our spam problem, the training sample consists of a set of email examples along with their associated labels. The training sample varies for different learning scenarios.

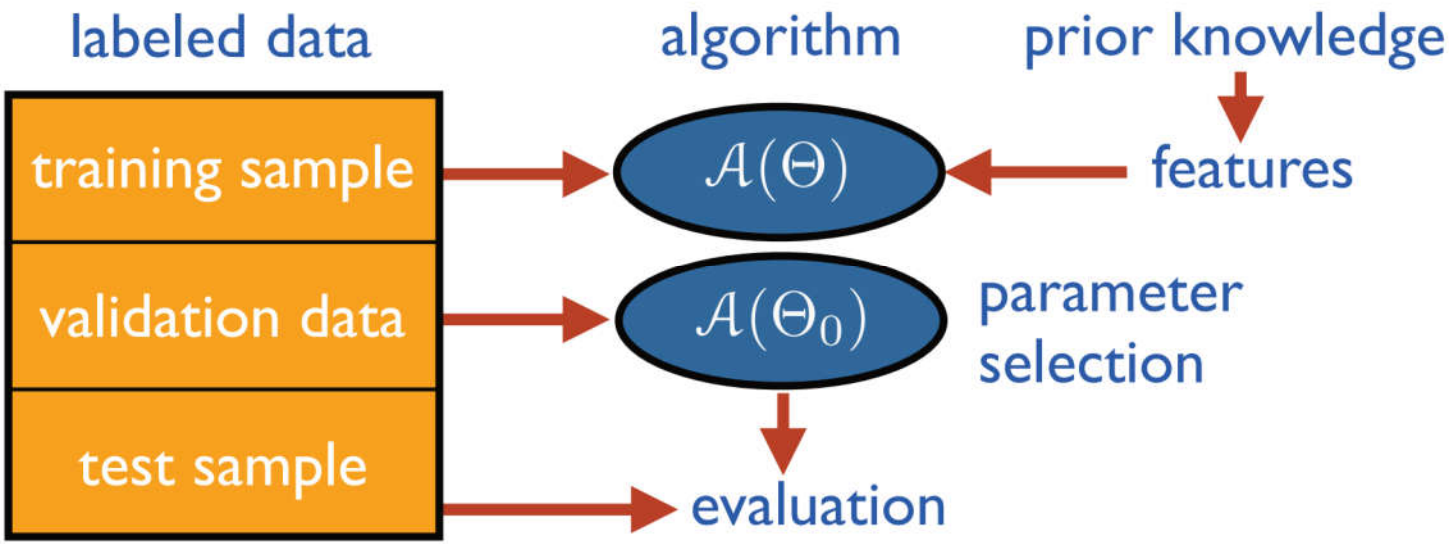
• *Validation sample*: Examples used to tune the parameters of a learning algorithm when working with labeled data. The validation sample is used to select appropriate values for the learning algorithm's free parameters (**hyperparameters**).

• *Test sample*: Examples used to evaluate the performance of a learning algorithm. The test sample is separate from the training and validation data and is not made available in the learning stage. In the spam problem, the test sample consists of a collection of email examples for which the learning algorithm must predict labels based on features. These predictions are then compared with the labels of the test sample to measure the performance of the algorithm.

• *Loss function*: A function that measures the difference, or loss, between a predicted label and a true label. Denoting the set of all labels as Y and the set of possible predictions as Y', a loss function L is a mapping L: $Y \times Y' \rightarrow R+$.

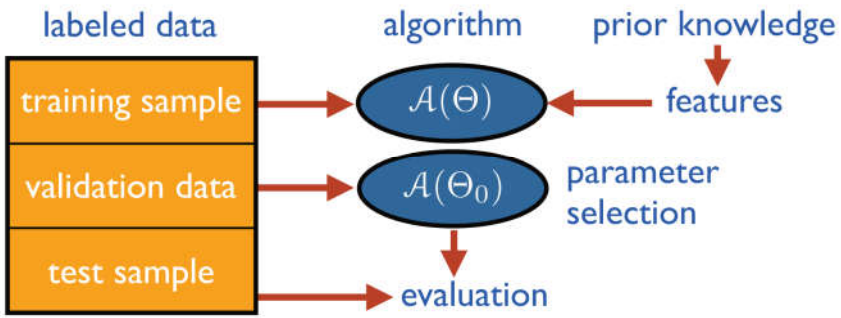
Learning Stages

• *Hypothesis set*: A set of functions mapping features (feature vectors) **X** to the set of labels **Y**. In our example, these may be a set of functions mapping email features to Y = {spam; non-spam}.



Learning Stages

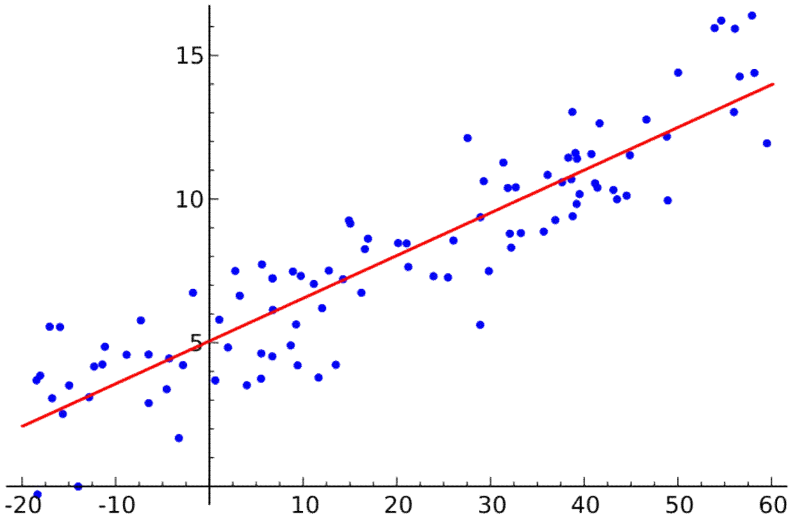
- We first randomly partition the data into a training sample, a validation sample, and a test sample.
- The size of each of these samples depends on a number of different considerations. For example, the amount of data reserved for validation depends on the number of hyperparameters of the algorithm. Also, when the labeled sample is relatively small, the amount of training data is often chosen to be larger than that of the test data.
- Next, we associate relevant features to the examples. This is a critical step in the design of machine learning solutions. Useful features can effectively guide the learning algorithm, while poor or uninformative ones can be misleading.
- Now, we use the features selected to train our learning algorithm by tuning the values of its free parameters (also called *hyperparameters*). For each value of these parameters, the algorithm selects a different hypothesis out of the hypothesis set. We choose the one resulting in the best performance on the validation sample.
- Finally, using that hypothesis, we predict the labels of the examples in the test sample. The performance of the algorithm is evaluated by using the loss function associated to the task.
- Thus, the performance of an algorithm is of course evaluated based on its test error and not its error on the training sample

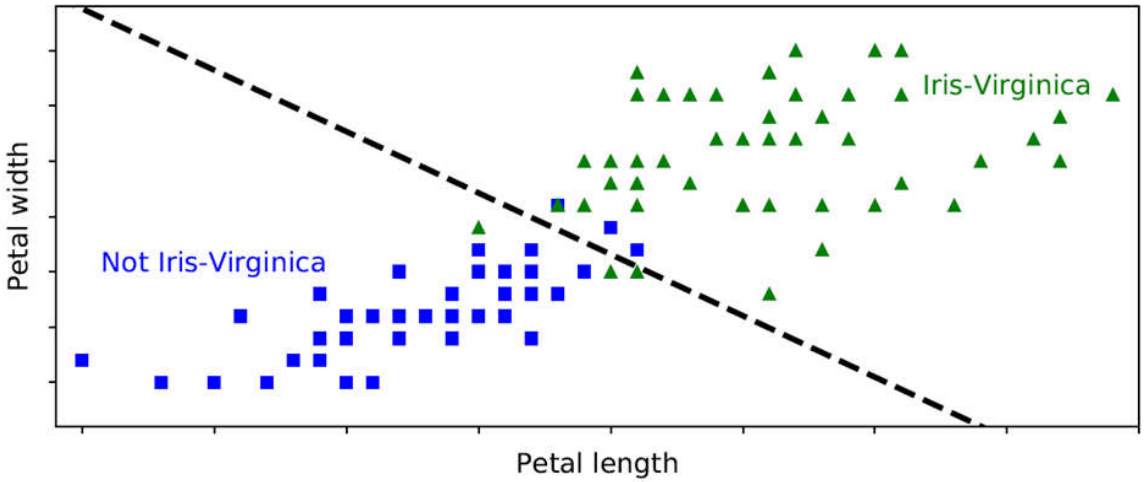


یادگیری نظارتی

۱ پیش بینی یک متغیر پیوسته (Regression)

۲ تخمین یک متغیر گسسته (Classification)





رگرسیون(برازش)

